CALDERWICK PUMP SYSTEMS LTD

Project Charter PRJ-11: Ardmoor Rig Control Upgrade

Project	PRJ-11
Version	1.1 (1 August 2025)
Sponsor	Chief Operating Officer
Project manager	Project Manager, Ardmoor Rig Upgrade
Budget approved by	Board of Directors, Resolution 2025-08 of 19 June 2025
Classification	Internal

1. Why this project

The hydraulic test rigs at Calderwick Pump Systems Ltd's Ardmoor plant run on control software and panels installed between 2012 and 2016. Spare parts are scarce, the firmware has known faults, and the old controls cannot be separated from the office network, which is a cyber security risk. The project replaces the control systems of all five rigs with a common design.

2. Scope

- New control engineering and integration for rigs TR-1, TR-2, TR-3, TR-4 and TR-5.
- New AC-HPU-7 PLC controllers on the hydraulic power units of all five rigs, the same controller family already used at Ardmoor; the controllers removed are kept as spares.
- New operator panels (HMI) on all five rigs.
- A separate, segmented OT network for the rigs with vendor access only through the jump host.
- Out of scope: mechanical work on the rigs and the test hall building.

Rig TR-5 was transferred from Hollins Cross, mechanically refurbished and entered service at Ardmoor on 29 September 2025 with its old controls; it receives the new controls in March 2026.

3. Budget

Table 1 - Budget

Item	Budget (USD)
Control engineering and integration (Ardent Controls Ltd, tender TE-2025-03)	192,500
PLC controllers AC-HPU-7, 5 units	162,000
Operator panels AC-HMI-12, 5 units	36,500
OT network and security	41,000
Contingency	48,000

Item	Budget (USD)
Total	480,000

4. Milestones

Table 2 - Milestones

Milestone	Date
Design freeze	30 September 2025
TR-1 cut-over	3 to 7 November 2025
TR-2 cut-over	1 to 5 December 2025
TR-3 cut-over	12 to 16 January 2026
TR-4 cut-over	9 to 13 February 2026
TR-5 cut-over	9 to 13 March 2026
Project closure	31 March 2026

Each rig may be out of service for at most one week. The Plant Manager, Ardmoor reschedules customer tests around the cut-overs.

5. Roles

Table 3 - Project roles

Role in project	Calderwick role	Person
Sponsor	Chief Operating Officer	Adaeze Okafor
Project manager	Project Manager, Ardmoor Rig Upgrade	Ishaan Kapoor
Technical lead	OT Systems Engineer	Elif Demir
Security adviser	Information Security Officer	Yusuf Karimi
Site lead	Plant Manager, Ardmoor (P2)	Ragnar Holt

6. Risks and change control

The main risks are R-07 (single maintenance supplier for the rigs) and R-11 (cyber attack on plant control networks); see the Risk Register. Changes that raise the budget by more than 5% need the sponsor's approval; more than 10% needs a new Board resolution.